IGCSE 0607 Practice Questions — E10.4 — Combined Questions & Markscheme

Averages and measures of spread

Total Questions: 15 — **Total Marks:** 105 Marks

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1 A student records the number of hours she spends reading each week for seven weeks.
The results are:

7 4 8 4 12 9 5

(a) **(a)** Write down the mode.

..... **[1]**

(b) Find the median.

..... **[1]**

(c) Work out the range.

..... **[1]**

2 The number of goals scored by a football team in each of their last 10 matches is shown below.

2 0 1 4 1 0 3 1 2 1

(a) Find the mode.

..... **[1]**

(b) Find the median.

..... **[2]**

(c) Calculate the mean.

..... **[2]**

3 A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

(a) Find the median.

..... **[1]**

(b) Find the range.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

(d) Calculate the mean.

..... **[2]**

4 The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

(a) Write down the value of x.

$x =$ ..... **[1]**

(b) Find the range of the number of points.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Calculate the mean.

..... **[2]**

(e) In the 8th game, the player scores y points.
The new mean number of points for all 8 games is 9.5.
Find the value of y.

$y =$ ..... **[2]**

5 The waiting times, in minutes, for 11 patients at a dental clinic are listed below.

15 11 23 8 50 14 18 10 15 21 13

(a) Write down the mode.

..... minutes **[1]**

(b) Find

(i) the median,

..... minutes **[2]**

(ii) the interquartile range.

..... minutes **[2]**

(c) Calculate the mean waiting time.

..... minutes **[2]**

(d) Explain why the median is a better measure of average than the mean for these data.

..... **[1]**

6 A botanist measures the heights, h cm, of 50 plants in a greenhouse.
The grouped frequency table shows the results, where x and y are positive integers.

Height (h cm)	Frequency
$0 < h \leq 10$	4
$10 < h \leq 20$	x
$20 < h \leq 30$	18
$30 < h \leq 40$	y
$40 < h \leq 50$	6

(a) Use the total number of plants to show that $x + y = 22$.

[1]

(b) The estimated mean height of the 50 plants is 27 cm.

(i) Show that $3x + 7y = 122$.

[3]

(ii) Solve the simultaneous equations $x + y = 22$ and $3x + 7y = 122$ to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(c) Write down the modal class of the distribution.

..... **[1]**

(d) Ten more plants are measured, and all ten have heights in the interval $30 < h \leq 40$.

(i) State the new modal class.

..... **[1]**

(ii) Calculate the new estimated mean height of all 60 plants, giving your answer as a mixed number.

..... cm **[3]**

7 The table shows the heights, h cm, of 80 plants.

Height (h cm)	**Frequency**
$10 < h \leq 20$	12
$20 < h \leq 30$	25
$30 < h \leq 40$	28
$40 < h \leq 50$	11
$50 < h \leq 60$	4

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

8 The table shows information about the heights, h cm, of 50 plants.

Height (h cm)	$0 < h \leq 10$	$10 < h \leq 20$	$20 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 50$
Frequency	4	12	20	10	4

(a) **(a)** Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

9 A survey was conducted on the masses, m grams, of 80 apples. The results are shown in the frequency table.

Mass (m grams)	$80 < m \leq 90$	$90 < m \leq 100$	$100 < m \leq 110$	$110 < m \leq 120$	$120 < m \leq 130$
Frequency	8	22	31	14	5

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the apples.

..... g **[4]**

(c) Explain why your answer to part (b) is an estimate.

.....

..... **[2]**

10 The table shows information about the time, t seconds, taken by 100 students to complete a puzzle.

Time (t seconds)	Frequency
$10 < t \leq 20$	14
$20 < t \leq 30$	28
$30 < t \leq 40$	32
$40 < t \leq 50$	18
$50 < t \leq 60$	8

(a) Write down the modal time interval.

..... **[1]**

(b) Calculate an estimate of the mean time.

..... seconds **[4]**

(c) Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

(d) Write down the interval that contains the upper quartile.

..... **[1]**

11 The table shows information about the masses, m grams, of 80 apples harvested from an orchard.

Mass (m grams)	Frequency
$100 \leq m < 120$	10
$120 \leq m < 140$	22
$140 \leq m < 160$	28
$160 \leq m < 180$	14
$180 \leq m < 200$	6

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the 80 apples.

..... g **[4]**

(c) A second crate contains 40 apples with a mean mass of 155 g. Calculate an estimate of the mean mass of all 120 apples combined.

..... g **[3]**

(d) State the class interval that contains the median mass of the 80 apples.

..... **[1]**

12 A factory produces 60 items.
The masses, m kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

Mass (m kg)	**Frequency**
$0 < m \leq 20$	p
$20 < m \leq 40$	14
$40 < m \leq 60$	16
$60 < m \leq 80$	q
$80 < m \leq 100$	5

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[5]**

(b) State the class interval which contains the median mass.
You must show your working.

..... **[2]**

13 The mean of five numbers is 12.
The median of the five numbers is 11.
The mode of the five numbers is 8.
The range of the five numbers is 15.

(a) Find the five numbers.

..... **[5]**

(b) A sixth number, k, is added to the list.
The new mean is 13.
Work out the value of k.

$k =$ ..... **[2]**

14 Six numbers are given in terms of x.

$$x-3 \qquad x \qquad x+2 \qquad 2x \qquad 2x+5 \qquad 3x$$

The mean of these six numbers is 9.

(a) **(a)** Calculate the value of x.

$x =$ ..... **[3]**

(b) Find the interquartile range of the six numbers.

..... **[2]**

(c) A seventh number, y, is added to the original six numbers.
The new median of the seven numbers is 8.
Find the value of y.

$y =$ ..... **[2]**

15 The table shows information about the masses, m kg, of 50 parcels.

Mass (m kg)	Frequency
$0 < m \leq 2$	15
$2 < m \leq 4$	x
$4 < m \leq 6$	y
$6 < m \leq 8$	8
$8 < m \leq 10$	2

The estimate of the mean mass of the parcels is 3.8 kg.

(a) Use the total number of parcels to write down an equation in x and y, and show that it simplifies to $x + y = 25$.

[1]

(b) Use the estimate of the mean to show that $3x + 5y = 101$.

[4]

(c) Solve the simultaneous equations to find the value of x and the value of y.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(d) Write down the interval that contains the median mass.

..... **[1]**

1 **(a)** **(a)** 4 **[B1]**

(b) Orders the data: $4, 4, 5, 7, 8, 9, 12$ **[M1]**

7 **[A1]**

(c) $12 - 4 = 8$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 1 **[B1]**

(b) Ordered list: $0, 0, 1, 1, 1, 1, 2, 2, 3, 4$ **[M1]**

1 **[A1]**

(c) $\dfrac{2+0+1+4+1+0+3+1+2+1}{10} = \dfrac{15}{10}$ **[M1]**

1.5 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** 16 **[B1]**

(b) 21 **[B1]**

(c) Ordered list seen: $5, 8, 10, 13, 14, 16, 18, 20, 22, 24, 26$ **[M1]**

12 **[A1]**

(d) $(14 + 20 + 8 + 24 + 16 + 5 + 26 + 13 + 22 + 10 + 18) \div 11$ or $176 \div 11$ **[M1]**

16 **[A1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** 7 **[B1]**

(b) $12 - 7 = 5$ **[B1]**

(c) 8 **[B1]**

(d) $(7 + 7 + 7 + 8 + 10 + 12 + 12) \div 7$ or $\dfrac{63}{7}$ **[M1]**

9 **[A1]**

(e) $\dfrac{63 + y}{8} = 9.5$ or 9.5×8 **[M1]**

$76 - 63 = 13$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** 15 **[B1]**

(b) **(i)** Ordered list: $8, 10, 11, 13, 14, 15, 15, 18, 21, 23, 50$ **[M1]**

15 **[A1]**

(ii) Lower quartile $Q_1 = 11$ and upper quartile $Q_3 = 21$ **[M1]**

$21 - 11 = 10$ **[A1]**

(c) $\dfrac{8+10+11+13+14+15+15+18+21+23+50}{11} = \dfrac{198}{11}$ **[M1]**

18 **[A1]**

(d) The value 50 is an extreme outlier that increases the mean, whereas the median is not affected by extreme values **[B1]**

[Back to Question] *[Total Marks: 8]*

6 **(a)** $4 + x + 18 + y + 6 = 50$ simplifying to $x + y = 22$ **[B1]**

(b) **(i)** $\sum fm = 4(5) + 15x + 18(25) + 35y + 6(45) = 15x + 35y + 740$ **[M1]**

$\dfrac{15x + 35y + 740}{50} = 27$ giving $15x + 35y + 740 = 1350$ **[M1]**

$15x + 35y = 610$ dividing by 5 to give $3x + 7y = 122$ **[A1]**

(ii) Eliminating one variable, for example $(3x + 7y) - 3(x + y) = 122 - 66$ giving $4y = 56$ **[M1]**

$y = 14$ **[A1]**

$x = 8$ **[A1]**

(c) $20 < h \leq 30$ **[B1]**

(d) **(i)** $30 < h \leq 40$ **[B1]**

(ii) New $\sum fm = 1350 + 10(35) = 1700$ **[M1]**

$\dfrac{1700}{60} = \dfrac{85}{3}$ **[M1]**

$28\dfrac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 12]*

7 **(a)** $30 < h \leq 40$ **[B1]**

(b) Uses midpoints $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = (15 \times 12) + (25 \times 25) + (35 \times 28) + (45 \times 11) + (55 \times 4)$ or
$180 + 625 + 980 + 495 + 220 = 2500$ **[M1]**

$\dfrac{2500}{80}$ **[M1]**

31.25 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** **(a)** $20 < h \leq 30$ **[B1]**

(b) Uses midpoints: $5, 15, 25, 35, 45$ **[M1]**

$\Sigma fx = (5 \times 4) + (15 \times 12) + (25 \times 20) + (35 \times 10) + (45 \times 4)$ **[M1]**

$\Sigma fx = 20 + 180 + 500 + 350 + 180 = 1230$ **[M1]**

$\frac{1230}{50} = 24.6$ **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** $100 < m \leq 110$ **[B1]**

(b) Midpoints used: $85, 95, 105, 115, 125$ **[M1]**

$\Sigma fx = 85 \times 8 + 95 \times 22 + 105 \times 31 + 115 \times 14 + 125 \times 5$ **[M1]**

$\Sigma fx = 680 + 2090 + 3255 + 1610 + 625 = 8260$ **[M1]**

$\text{Mean} = \frac{8260}{80} = 103.25$ **[A1]**

(c) The exact masses of individual apples are not known **[B1]**

The calculation uses the midpoints of the class intervals, assuming the data is evenly distributed within each class **[B1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** $30 < t \leq 40$ **[B1]**

(b) Midpoints: $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = 15(14) + 25(28) + 35(32) + 45(18) + 55(8)$ **[M1]**

$= 210 + 700 + 1120 + 810 + 440 = 3280$ **[M1]**

$\frac{3280}{100} = 32.8$ **[A1]**

(c) $18 + 8 = 26$ **[M1]**

26 **[A1]**

(d) $40 < t \leq 50$ **[B1]**

[Back to Question] *[Total Marks: 8]*

11 **(a)** $140 \leq m < 160$ **[B1]**

(b) Mid-interval values: $110, 130, 150, 170, 190$ **[B1]**

$\sum fm = 10(110) + 22(130) + 28(150) + 14(170) + 6(190) = 11680$ **[M1]**

$\frac{11680}{80}$ **[M1]**

146 **[A1]**

(c) Total mass of second crate $= 40 \times 155 = 6200$ **[M1]**

$\frac{11680 + 6200}{120} = \frac{17880}{120}$ **[M1]**

149 **[A1]**

(d) $140 \leq m < 160$ **[B1]**

[Back to Question] *[Total Marks: 9]*

12 **(a)** $p + q = 25$ **[B1]**

$\dfrac{10p + 420 + 800 + 70q + 450}{60} = 48$ or $10p + 70q = 1210$ [B2] (Allow B1 for unsimplified $\Sigma fx = 10p + 420 + 800 + 70q + 450$ seen)

Correctly sets up and solves simultaneous equations to eliminate one variable **[M1]**

$p = 9$, $q = 16$ **[A1]**

(b) Identifies median is between 30th and 31st value and shows cumulative frequencies 23 and 39 (or $9 + 14 = 23$ and $23 + 16 = 39$) **[M1]**

$40 < m \leq 60$ **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** Since the mode is 8, there must be at least two 8s. **[M1]**

Since the median is 11, the middle number when ordered is 11. The first three numbers are $8, 8, 11$. **[M1]**

The mean is 12, so the sum of the five numbers is $5 \times 12 = 60$. **[M1]**

The range is 15, so the largest number is $8 + 15 = 23$. **[M1]**

Let the fourth number be x. Then $8 + 8 + 11 + x + 23 = 60 \implies 50 + x = 60 \implies x = 10$. **[A1]**

(b) New sum $= 6 \times 13 = 78$ **[M1]**

$k = 78 - 60 = 18$ **[A1]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** **(a)** Sets up equation for the mean: $\dfrac{(x-3) + x + (x+2) + 2x + (2x+5) + 3x}{6} = 9$ **[M1]**

Simplifies numerator: $10x + 4 = 54$ **[M1]**

$10x = 50 \implies x = 5$ **[A1]**

(b) Substitutes $x = 5$ to find the ordered list: $2, 5, 7, 10, 15, 15$ **[M1]**

Lower quartile $= 5$ and Upper quartile $= 15$, so IQR $= 15 - 5 = 10$ **[A1]**

(c) Recognises that for 7 numbers, the median is the 4th value in the ordered list. **[M1]**

The current values are $2, 5, 7, 10, 15, 15$. For the 4th value to be 8, y must be placed exactly at the 4th position, giving $2, 5, 7, 8, 10, 15, 15$. Thus $y = 8$. **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $15 + x + y + 8 + 2 = 50$ leading to $x + y = 25$ **[B1]**

(b) Midpoints: $1, 3, 5, 7, 9$ **[M1]**

$$\frac{1(15)+3(x)+5(y)+7(8)+9(2)}{50}=3.8 \quad \textbf{[M1]}$$

$15+3x+5y+56+18=190$ **[M1]**

$3x+5y+89=190$ leading to $3x+5y=101$ **[A1]**

(c) $3x+3y=75$ or substitution method $3(25-y)+5y=101$ **[M1]**

$2y=26 \implies y=13$ **[A1]**

$x=12$ **[A1]**

(d) $2<m\leq 4$ **[B1]**

[Back to Question] *[Total Marks: 9]*